EV Battery Course — Module 8: Battery Thermal Management

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Li-ion cells work best in a narrow band of roughly 20–40 °C: colder and they lose power and cannot charge safely; hotter and they age fast and move closer to thermal runaway. The thermal system keeps them there, and in India that mostly means removing heat. This module covers heat generation, cooling and heating technologies, thermal runaway containment, and a worked thermal calculation on the 60 kWh pack from Module 6.

1. Learning objectives

- Explain where heat comes from inside a cell during charge and discharge.
- Explain why temperature gradients matter as much as average temperature.
- Compare air, liquid, refrigerant and immersion cooling.
- Describe cold-climate heating and cold-start limits.
- Describe the hardware of a liquid thermal system.
- Explain thermal runaway, propagation, and how packs are designed to contain it.
- Calculate heat load, coolant flow, temperature rise and heating time for a pack.

2. Fundamentals: temperature and the battery

Temperature	Effect
Below 0 °C	Ion transport and charge transfer slow sharply; charging risks lithium plating; power falls
0–15 °C	Reduced power and charge rate; range loss
20–40 °C	Best balance of performance and life
40–55 °C	Faster SEI growth and electrolyte breakdown; ageing accelerates
Above ~60–80 °C	SEI breaks down; exothermic side reactions start; risk of runaway rising
~130 °C	PE separator melts (shutdown); beyond that, separator collapse and internal short

Temperature	Effect
~200–270 °C	Cathode decomposition (lower for nickel-rich, higher for LFP); thermal runaway

Also critical: **uniformity.** Cells at different temperatures age at different rates, so a pack with a 10 °C spread will drift out of balance. A common design target is a cell-to-cell spread of under about 5 °C.

3. Heat generation

Cell heat has two parts:

$$\dot{Q} = I\,(OCV - V_t) \; - \; I\,T\,\frac{dOCV}{dT}$$

- **Irreversible heat (first term):** Joule (ohmic) heating plus polarisation losses. Always positive. Grows with I^2. Dominates at high current.
- **Reversible (entropic) heat (second term):** from the entropy change of the reaction; it can heat or cool the cell depending on SOC and current direction. Significant only at low currents.

I in A (positive for discharge in this convention), T in K, dOCV/dT in V/K.

A simpler engineering approximation at high current: $Q \approx I^2 R_{eff}$, where R_{eff} includes polarisation (often 1.2–1.5× DCIR).

Gradients inside cells. Heat is generated throughout the electrode stack but removed only at the surface. Large cells and cooling on a single face create core-to-surface gradients; tabs and busbars are also hot spots, because current concentrates there.

4. Cooling and heating technologies

Method	How it works	Heat-removal capability	Pros	Cons	Examples
Passive / natural air	Conduction to casing, natural convection	Very low	Cheapest, no parasitic power	Not suitable for fast charge or hot climate	Many low-cost 2W/3W packs
Forced air	Fans blow cabin or ambient air through gaps	Low	Simple, light	Low heat capacity of air, big ducts, gradients	Early Nissan Leaf (passive), Toyota hybrids (forced air)

Method	How it works	Heat-removal capability	Pros	Cons	Examples
Liquid (indirect)	Water-glycol through cold plates under or between cells	High	Uniform, compact, connects to chiller and heater	Leak risk, pump, cost, mass	Most modern EVs: Tata, Hyundai, VW, Tesla
Refrigerant (direct)	Refrigerant evaporates in plates under cells	Very high	No secondary loop; strong cooling	Complex control; non-uniform evaporation	Some BMW i3 versions, some premium models
Immersion	Cells submerged in dielectric fluid	Highest, most uniform	Excellent propagation resistance, fast charging	Fluid mass and cost, sealing, servicing	Some performance cars, buses and storage; growing interest

Heating. PTC heaters in the coolant loop, heat pumps that also serve the cabin, motor/inverter waste heat routed to the battery, and internal heating through controlled current pulses (used by some makers). Preconditioning before DC charging uses navigation to warm the pack to the charging window on arrival.

5. Diagram: a typical liquid thermal loop

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flowchart LR
  PUMP["Electric pump"] --> HEATER["PTC heater (cold)"] --> PACK["Battery
cold plates"] --> VALVE["Multi-way valve"]
  VALVE --> CHILLER["Chiller (heat to AC refrigerant)"] --> PUMP
  VALVE --> RAD["Front radiator (mild conditions)"] --> PUMP
  BMS["BMS temperature requests"] -.-> PUMP
  BMS -.-> VALVE
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Components: cold plates (brazed aluminium channels or stamped plates), thermal interface material (TIM: gap pads, gap fillers or thermally conductive adhesive) between cells and plate, pump, valves, PTC heater, chiller (plate heat exchanger to the AC circuit), radiator, coolant temperature sensors, and expansion tank.

6. Step-by-step: what happens during a 30-minute DC fast charge in summer

1. Navigation sees a fast-charge stop and preconditions the pack to about 30 °C.
2. Charging starts at high current; I^2R heat rises sharply.

3. BMS requests maximum cooling; the valve routes coolant to the chiller; the AC compressor runs hard.
4. Heat flows: electrode → cell can → TIM → cold plate wall → coolant → chiller → refrigerant → condenser → ambient air.
5. If the hottest cell nears about 45–50 °C, the BMS lowers charging current (one reason for charging taper).
6. After charging, the cooling continues briefly to bring the pack back toward its target.

7. Engineering equations

$$\dot{Q}_{gen} \approx I^2 R_{eff}$$

Temperature rise without cooling (lumped mass):

$$\Delta T = \frac{\dot{Q}\,t}{m\,c_p}$$

Coolant flow and coolant temperature rise:

$$\dot{m} = \frac{\dot{Q}}{c_p\,\Delta T_{coolant}}$$

Conduction through a layer (TIM, wall):

$$R_{th} = \frac{t}{k\,A} \qquad \Delta T = \dot{Q}\,R_{th}$$

Convection to coolant:

$$\dot{Q} = h\,A\,(T_{wall} - T_{coolant})$$

Pump power:

$$P_{pump} = \frac{\Delta p\,\dot{V}}{\eta_{pump}}$$

Chiller compressor power:

$$P_{comp} = \frac{\dot{Q}}{COP}$$

Symbols: m = mass (kg), c_p = specific heat (J/(kg·K)), t = time (s) or thickness (m), k = thermal conductivity (W/(m·K)), A = area (m²), h = heat transfer coefficient (W/(m²·K)), Δp = pressure drop (Pa), $\dot{V}$ = volumetric flow (m³/s), COP = coefficient of performance.

8. Worked thermal calculation: the 60 kWh pack under high power

Pack (from Module 6): 120S1P LFP, 384 V, R_pack = 40 mΩ, cell mass 351 kg, plate footprint about 1.7 × 1.3 m.

Condition: 100 kW sustained (fast charge, or a long uphill at high speed), 35 °C ambient.

Assumptions: R_eff = 1.3 × R_pack (polarisation); cell c_p = 1,000 J/(kg·K); effective cell-to-plate contact area 1.33 m²; TIM 1 mm thick, k = 2 W/(m·K); water-glycol c_p = 3,500 J/(kg·K), density 1.06 kg/L; flow 12 L/min; loop pressure drop 40 kPa; pump efficiency 40%; chiller COP 2.5.

Step	Calculation	Result
Current	100,000 ÷ 384	260.4 A
Heat generated	$260.4^2 \times 0.040 \times 1.3$	**3.53 kW**
Temperature rise in 10 min with no cooling	3,526 × 600 ÷ (351 × 1,000)	**6.0 K** (30 min → 18 K)
Coolant mass flow	12 ÷ 60 × 1.06	0.212 kg/s
Coolant temperature rise	3,500 ÷ (0.212 × 3,500)	**4.7 K** inlet to outlet
TIM thermal resistance	0.001 ÷ (2 × 1.33)	0.00038 K/W
ΔT across TIM	3,500 × 0.00038	**1.3 K**
Pump power	40,000 × 0.0002 ÷ 0.4	**20 W**
Chiller compressor power	3,500 ÷ 2.5	**1.4 kW**

Interpretation. With coolant entering at about 25 °C, the outlet is about 30 °C, the plate wall a little warmer, the cell base about 1.3 K above that, and the cell core several K above the base. Hottest cells likely sit around 35–40 °C: acceptable. The chiller's 1.4 kW is real parasitic energy, which matters for charging efficiency in hot weather.

Why not air? Removing 3.5 kW with a 10 K air temperature rise needs 3,500 ÷ (1,005 × 10) = 0.35 kg/s ≈ **1,045 m³/h** of air through the pack: impractical ducts and fan noise.

Cold-climate case. Heating the 351 kg of cells from −5 °C to 15 °C needs 351,000 × 20 = 7.02 MJ = **1.95 kWh**; with a 5 kW heater and no losses, about **23 minutes**. That is why preconditioning matters before fast charging in winter.

Limitations. Lumped-mass model only; real designs need CFD and 3D thermal FEA, cell-level heat maps from calorimetry, transient drive-cycle analysis, ageing (resistance growth increases heat), and validation on hardware.

9. Thermal runaway and propagation

Runaway sequence (simplified): SEI breakdown (~80–120 °C) → anode reacts with electrolyte → separator melts (~130–170 °C) → internal short → cathode decomposition releases oxygen (NMC ~200 °C; LFP ~250–270 °C) → electrolyte combustion, temperatures of several hundred °C, and large volumes of flammable and toxic gas.

Propagation: heat from one failing cell, hot gases and burning ejecta can trigger neighbours, cascading through the pack.

Design measures

Layer	Measures
Prevent	Cell quality, BMS limits, thermal control, crash structure
Detect	Cell voltage drop, rapid temperature rise, pack pressure sensors, gas (H_2, CO, VOC) sensors
Contain	Thermal barriers between cells (mica, aerogel, ceramic paper), spacing, cooling plates as heat sinks
Vent	Directed vent channels and burst valves to carry gas and ejecta away from cabin and neighbours
Warn	Occupant warning in time to exit

Regulation: China's GB 38031-2020 requires no fire or explosion for 5 minutes after a single cell's thermal runaway, with a warning to occupants. India's AIS-156 (two- and three-wheelers) and AIS-038 Rev 2 (cars) were amended in 2022–23 to add thermal propagation and other safety requirements after a run of scooter fires. Check the current amendment text before use.

10. Trade-offs

Factor	Air	Liquid cold plate	Refrigerant direct	Immersion
Cooling effectiveness	Low	High	Very high	Very high
Temperature uniformity	Poor	Good	Fair	Excellent

Factor	Air	Liquid cold plate	Refrigerant direct	Immersion
Pressure drop / parasitic power	Fan power, low	Pump power, low	Compressor, medium	Pump power, higher (viscous fluid)
Packaging	Large ducts	Thin plates	Thin plates	Whole pack filled
Mass	Low	Medium	Medium	High
Cost	Low	Medium	Medium-high	High
Serviceability	Easy	Medium; leak checks	Needs refrigerant handling	Hard; drain and refill

11. Real-world EV examples

- **Tata Nexon EV, MG ZS EV:** liquid-cooled packs connected to the AC system, chosen for Indian summers.
- **Early Nissan Leaf:** passively cooled pack; faster degradation in hot climates such as Arizona was widely reported, a classic lesson.
- **Tesla:** serpentine cooling tube winding between cylindrical cells; heat pump in newer models.
- **Electric two-wheelers in India:** mostly air-cooled or passively cooled; Ather and some others add thermal pads and aluminium housings as heat spreaders.
- **CATL Qilin:** cooling plates between cells on their large faces, raising heat-exchange area.

12. Manufacturing applications

- **TIM dispensing:** gap filler bead volume and pattern are critical; voids under a cell mean a hot cell. Vision inspection, weight check, and sometimes ultrasonic or thermal imaging verify coverage.
- **Cold-plate brazing and leak testing:** helium or pressure-decay leak tests on every plate and the assembled loop.
- **Coolant fill:** vacuum fill to avoid air pockets; check for air-lock.
- **Thermal EOL test:** run coolant and current and check sensor readings are consistent.
- **Costing:** thermal system often 5–10% of pack cost; chiller and heater shared with HVAC.

13. Common mistakes and misconceptions

- **"Colder is always better."** Below about 15–20 °C power and charging suffer; target a band.
- **"Average temperature is enough."** Spread drives imbalance and local ageing.
- **"LFP doesn't need cooling."** It tolerates heat better but still ages faster when hot and needs cooling for fast charging.
- **"More TIM is better."** TIM is a resistance; thinner and well-filled is better. Excess TIM adds mass and cost.
- **"Thermal runaway can be stopped once started."** Within a cell, generally not; design focuses on preventing propagation.

14. Practical case study: TIM voids causing hot cells

Problem. EOL fast-charge tests show 2–3% of packs with one cell 8–10 °C hotter than its neighbours.

Investigation. Teardown finds incomplete gap-filler coverage under those cells. Root cause: dispensing nozzle partly blocked by cured material after line stoppages; bead weight drifted low with no alarm.

Correction. Add in-line bead-weight measurement with SPC limits, purge routine after stops, and vision inspection of the bead pattern before cell placement.

Lesson. Thermal performance depends on a manufacturing process with tight control, not only on design.

15. Revision notes

- Ideal operating band about 20–40 °C; keep cell spread under about 5 °C.
- Heat = irreversible (I^2R + polarisation) + reversible (entropic).
- Liquid cold plates dominate EVs; air is too weak for fast charging in hot climates.
- Preconditioning enables fast charging in cold weather.
- Runaway: prevent, detect, contain, vent, warn.

16. Glossary

Term	Meaning
TIM	Thermal interface material

Term	Meaning
Chiller	Heat exchanger moving heat from coolant to AC refrigerant
PTC heater	Self-regulating electric heater
COP	Coefficient of performance: heat moved per unit work
Preconditioning	Heating or cooling the pack before charging
Propagation	Spread of runaway from cell to cell
Entropic heat	Reversible heat from the reaction's entropy change

17. Knowledge check

Conceptual

1. Why does charging below 0 °C risk lithium plating?
2. Why does a temperature spread of 10 °C cause problems even if the average is fine?
3. Why is air cooling inadequate for fast-charging EVs in India?
4. What is the purpose of thermal barriers between cells?
5. Why does preconditioning shorten winter charging stops?

Numerical

1. A 2W pack (50 kg cells, c_p 1,000 J/(kg·K)) generates 300 W for 20 minutes with no cooling. What is the temperature rise?
2. Remove 5 kW with coolant rising 6 K (c_p 3,500). What mass flow is needed?
3. A TIM layer is 1.5 mm, k = 3 W/(m·K), area 0.8 m², heat 2 kW. What is the ΔT?

Application

1. An e-bus operator in Chennai sees faster capacity fade than expected. What thermal questions would you ask?
2. Your plant must choose between gap pads and dispensed gap filler. Compare for quality and cost.

18. Answer key

Conceptual

1. Lithium ions diffuse into graphite slowly when cold; at high current they accumulate at the surface and deposit as metallic lithium instead of intercalating.
2. Hotter cells age faster, losing capacity and gaining resistance, so the pack drifts out of balance and the weakest cell limits the pack sooner.
3. Air carries little heat per volume; removing several kW needs huge airflow, and in 40–45 °C ambient the air is already too warm.
4. To slow heat transfer from a failing cell to its neighbours, preventing or delaying propagation.
5. A warm pack has lower resistance and no plating risk, so the BMS allows high charging current from the start.

Numerical

1. 300 × 1,200 ÷ (50 × 1,000) = **7.2 K**.
2. 5,000 ÷ (3,500 × 6) = **0.238 kg/s** (≈ 13.5 L/min).
3. R = 0.0015 ÷ (3 × 0.8) = 0.000625 K/W → ΔT = 2,000 × 0.000625 = **1.25 K**.

Application

1. What are pack temperatures during depot charging and in service? Is cooling active while parked in the sun? What is the cell temperature spread? Is the chiller sized for 40+ °C ambient? Are charge rates temperature-limited? Are coolant flows and TIM coverage as designed?
2. Gap pads: easy to inspect and rework, no curing, but poorer conformity and higher thermal resistance, higher material cost per area. Gap filler: thinner and better contact, automated, lower material cost at volume, but needs dispensing control, curing time and inspection; rework is hard.

19. Further learning

- Pesaran, A., "Battery thermal management in EVs and HEVs: issues and solutions," NREL (classic paper).
- Bernardi, Pawlikowski & Newman, "A General Energy Balance for Battery Systems," *J. Electrochem. Soc.* 1985.
- GB 38031-2020 (China EV battery safety); AIS-156 and AIS-038 Rev 2 with amendments (India).
- NREL battery thermal publications (free).

Next: Module 9, charging technology.